

FUNCTIONAL ANALYSIS

(5 hours)

January 13, 2012

Please write on the front page your preferred time for the oral exam.

1. Let $C_b(\mathbb{R})$ be the space of all bounded continuous functions on $\mathbb{R}$, normed with

$$\|f\| = \sup_{x \in \mathbb{R}} |f(x)|.$$

- (i) Show that $(C_b(\mathbb{R}), \|\cdot\|)$ is a Banach space.
(ii) Show that the linear map $M : C_b(\mathbb{R}) \rightarrow l^\infty$ defined by

$$Mf = (f(n))_{n \geq 1}$$

is continuous and onto.

- (iii) Show that $(C_b(\mathbb{R}), \|\cdot\|)$ is not separable.

2. Let $f \in C([0, 1])$ and assume that there exists a positive constant C such that

$$\left| \int_0^1 p'(t)f(t)dt \right| \leq C\|p\|_2,$$

for all polynomials p , where $\|p\|_2^2 = \int_0^1 |p(t)|^2 dt$. Show that there exists a unique $g \in L^2([0, 1])$ such that

$$f(x) = \int_0^x g(t)dt, \quad \int_0^1 g(t)dt = 0.$$

3. Let X be a Banach space.

a) Prove that a linear map $M : X \rightarrow l^p$, $p \geq 1$, is continuous if and only if for every sequence (x_k) that converges in X to $x \in X$, we have that the n -th term of the sequence Mx_k converges to the n -th term of Mx for all n .

b) Prove that a linear map $M : X \rightarrow C([0, 1])$ is continuous if and only if for every $t \in [0, 1]$, the rule $x \rightarrow (Mx)(t)$ defines a continuous linear functional on X (Here $(Mx)(t)$ means the continuous function Mx evaluated at the point t).

c) State and prove a similar continuity criterion for linear maps $M : X \rightarrow Y$, where Y is an arbitrary Banach space.

4. Given a bounded sequence $\Lambda = (\lambda_n)$ in $\mathbb{C}$ consider the continuous linear map $M_\Lambda : l^2 \rightarrow l^2$ defined by

$$M_\Lambda(x_n) = (\lambda_n x_n).$$

a) Determine the spectrum of M_Λ . You can use without proof the fact that M_Λ is bounded if and only if Λ is bounded.

b) Characterize the sequences Λ for which M_Λ is compact.

5. Let X, Y be Banach spaces and consider the product space $X \times Y$ with the usual norm

$$\|(x, y)\| = \|x\|_X + \|y\|_Y.$$

Let $T : X \rightarrow X$, $S : Y \rightarrow Y$, be two bounded invertible linear maps and $R : Y \rightarrow X$ be bounded with norm

$$\|R\| < \min\left\{\frac{1}{\|T^{-1}\|}, \frac{1}{\|S^{-1}\|}\right\}.$$

Prove that the linear map $M : X \times Y \rightarrow X \times Y$ defined by

$$M(x, y) = (Tx + Ry, Sy),$$

is bounded and invertible.

6. Let H be a separable Hilbert space and let $T : H \rightarrow H$ be a symmetric bounded linear map ($T^* = T$).

a) Show that for every orthogonal projection P on H ($P^* = P$, $P^2 = P$) PTP is symmetric.

b) Prove that there exists a sequence C_n of compact symmetric linear maps such that $C_n x \rightarrow Tx$ for every $x \in H$.